



DATA-549 PROJECT REPORT

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Abstract

This project utilizes natural language processing techniques to predict outcomes of crowdfunding campaigns run through Kickstarter. It draws on text data from the project descriptions dating from 2017. By employing the techniques in this project, we were able to create models to predict whether a project was successfully able to be fully funded or if it failed.

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Introduction

The importance of crowdfunding mainly allows individuals access to capital and see if their idea is enough to garner validation that comes in the form of the community or the population it is trying to serve or represent. Mainly, crowdfunding initiatives are used in startups, social issue groups, and entrepreneurs. The gist of crowdfunding is to see if you can gain a following that will support what you are in essence selling whether it be for profit or non-profit.

This project will focus on seeing what drives backers to accept a project by way of text classification. To be able to break down the blurbs or descriptions of the proposed project and compare different vectorization techniques, clustering methods, and features sets to see which is best to accurately predict which projects would be accepted or “backed.”

Data Description

This dataset was found on Kaggle. It is a collection of Kickstarter project campaign descriptions run in 2017. It contains 215,513 rows each containing a unique description of the project and a binary state of the funding campaign. The data was collected from webrobots.io, who did the web scrapping. It was then cleaned and tidied to be used on Kaggle. There are no missing values in this dataset.

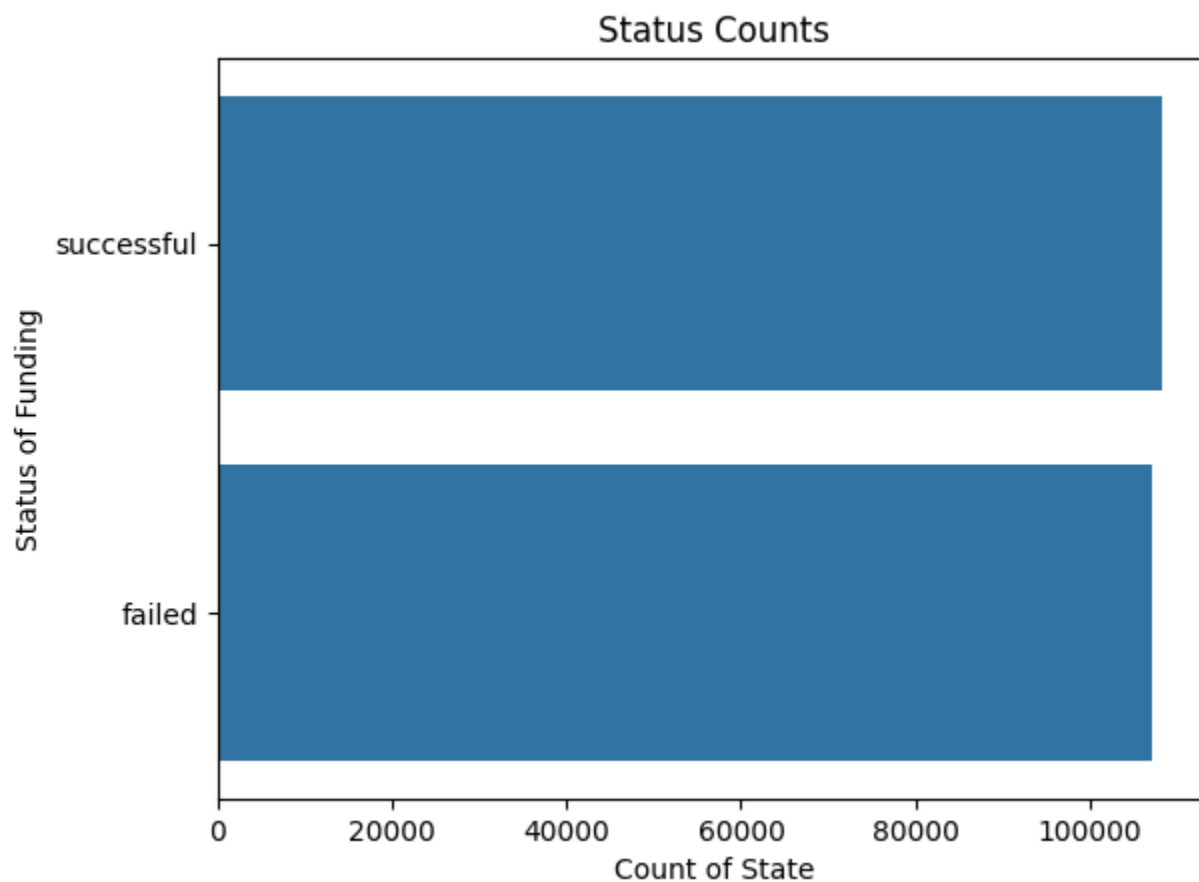
Column name	Type	Description
blurb	Character	Short description of the project
state	Character	Final state of the project

Exploratory Data Analysis

To get a better understanding of our data, we created a series of visualizations to explore the distribution of funding states and familiarize ourselves with the text data. The goal of

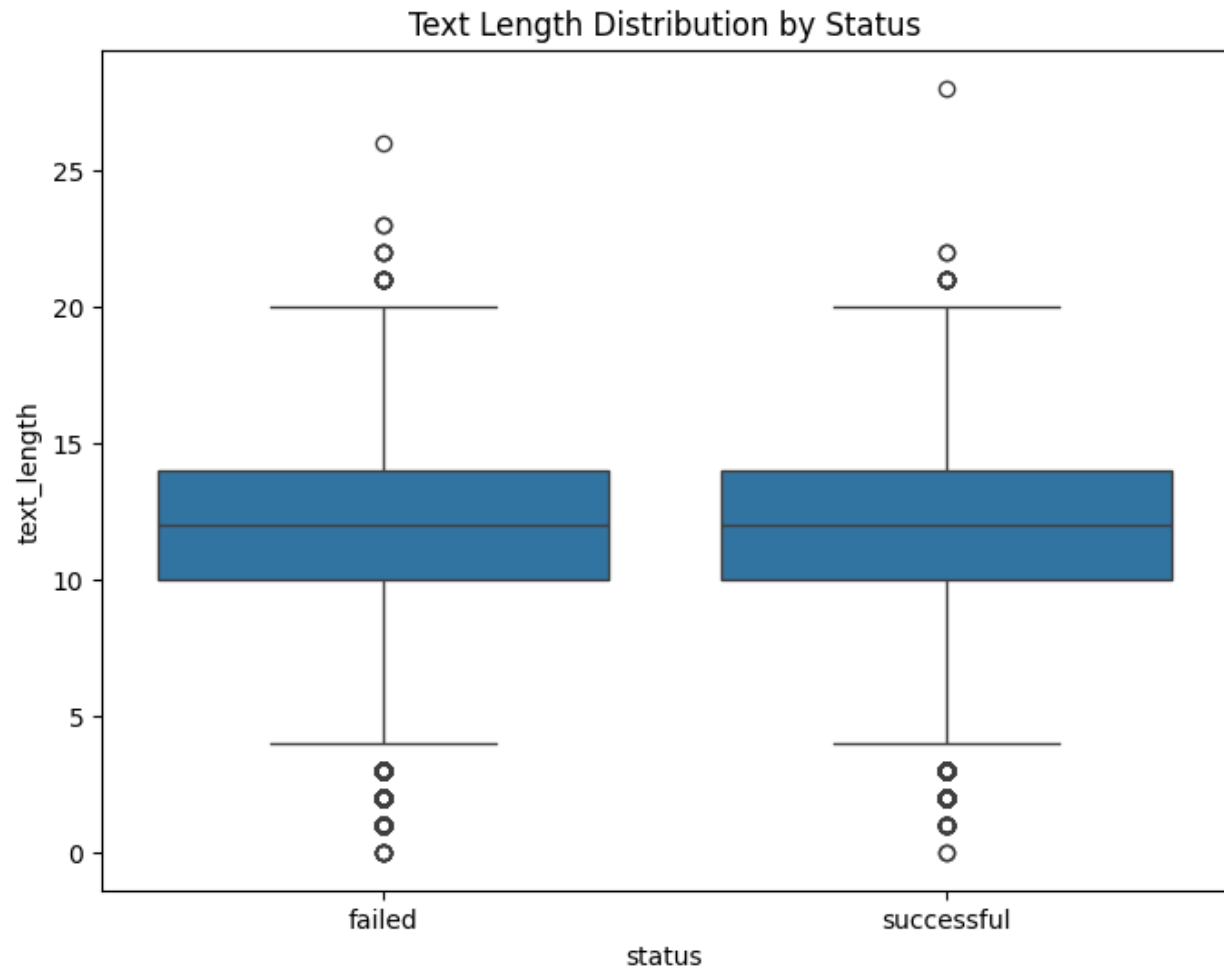
this analysis is to identify key applicant characteristics between funding states to assist us in the modeling process.

Figure 1: Distribution of Status



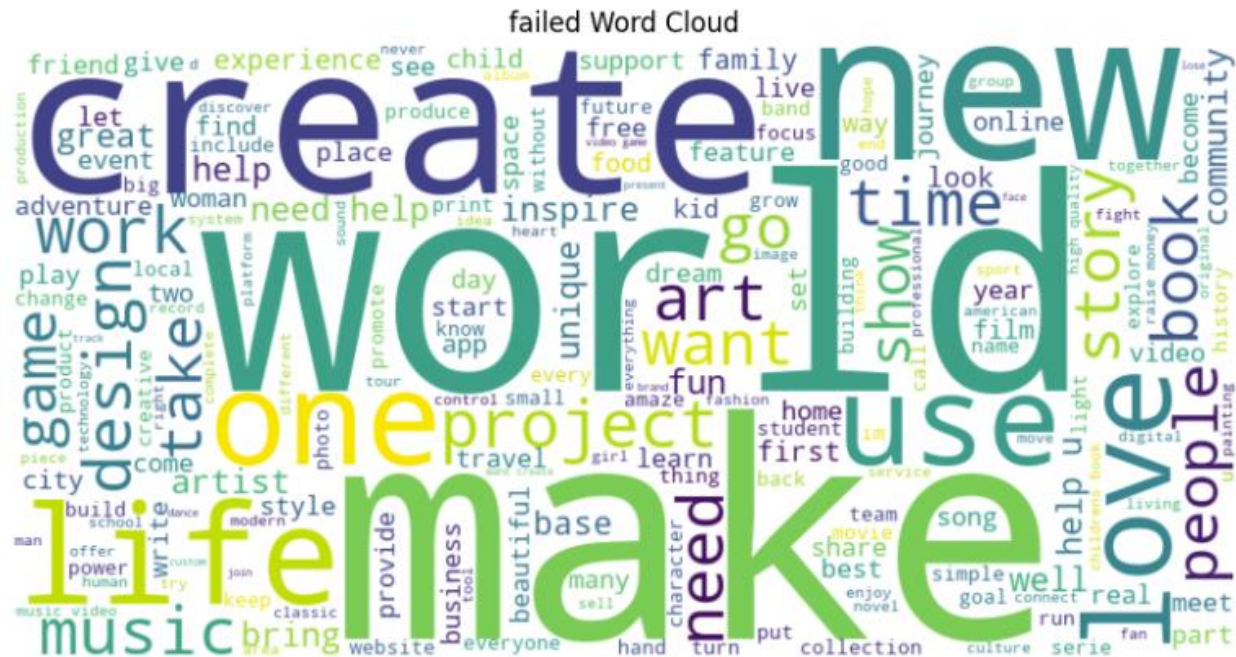
The above chart shows status counts of the Kickstarter funding campaigns. This dataset is balanced with a 50.25/49.75 split between successful and failed campaigns.

Figure 2: Length of Description Texts



The above chart shows the text length of each project description. From the chart we can see they are close to identical aside from a few outliers at the high end.

Figure 3: Failed Project Word Cloud



From the above word cloud, we can see the most frequent words in the project descriptions of failed campaigns. Some of the most common include world, make, create, and new.

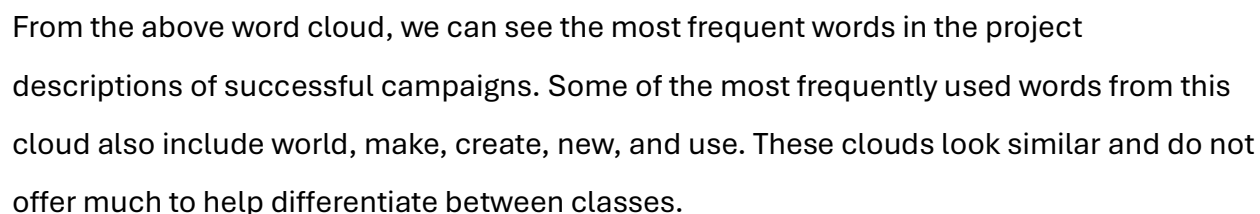
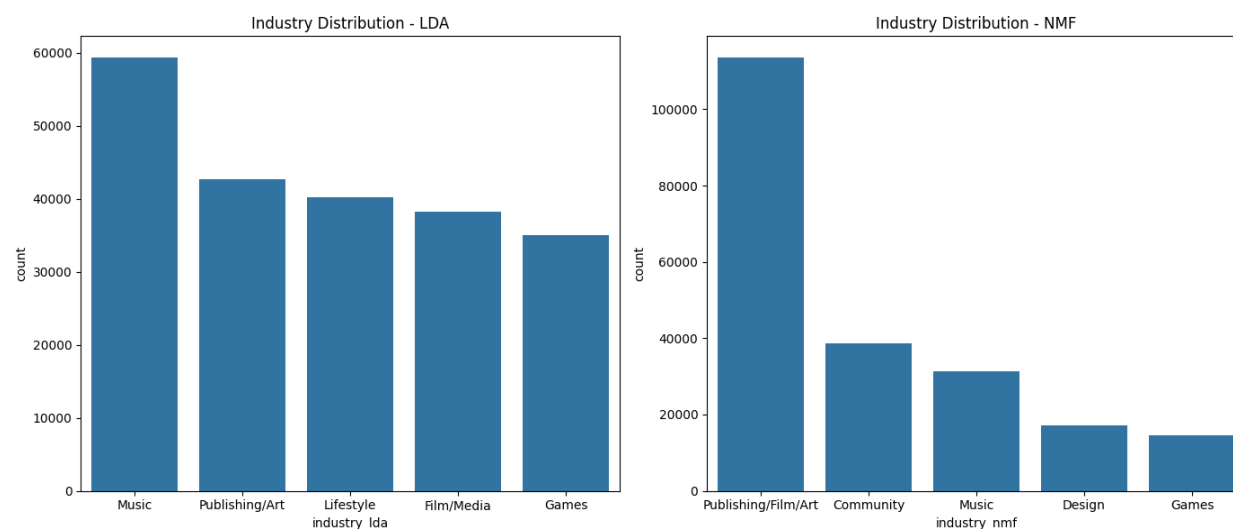


Figure 5: Industry Label Distributions



The above chart shows the distribution of project industries obtained from LDA and NMF topic models. We used 5 topics and assigned each topic an industry label that made the most sense from the top words in each topic. Music, publishing, film, and art were some of the most common industries we saw using this method on our dataset.

Preprocessing

Before building our models, we preprocessed the text data by removing the punctuation, stop words, pronouns, numbers, and performing lemmatization. We checked to see if there was any missing data and there was none. We dropped an extra column to set off our initial data frame.

After initial cleaning of punctuation, stopwords, pronouns, and numbers, the dataset was put through lemmatization as our method of text normalization. After which we renamed some columns to keep our data frame clean for the next step, which is base modeling.

Modeling

There were two vectorization techniques we considered in this process. One was Term Frequency-Inverse Document Frequency (TF-IDF) and BERT(Bidirectional Encoder Representations from Transformers transformation). TF-IDF was fitted with LDA (Latent Dirichlet Allocation) to extract top topics in the text. TF-IDF was also fitted with NMF (Non-Negative Matrix Factorization) for topic modeling. Both of which were combined with the data frame and mapped according to topic indices.

Apart from supervised methods, we investigated unsupervised clustering machine learning algorithms as well. Both KMeans and MiniBatchKMeans were fitted to BERT embeddings to compute cluster centroids. Both algorithms gave different results for cluster centroids.

Finally, there was a feature set bake-off that took part in determining which feature set representation was the best to predict success in our data, i.e., topics, embeddings, clusters, industries, or combinations.

Figure 6: Model Performance Summary

	Logistic	RandomForest	XGBoost	LightGBM	CatBoost	Set_Average
BERT_Industry_Label	0.649374	0.634582	0.649236	0.643303	0.659430	0.647185
bert	0.648980	0.634580	0.649585	0.643445	0.658445	0.647007
BERT_Industry_Label_nmf	0.649104	0.634337	0.649324	0.643079	0.659178	0.647004
BERT_Industry_Label_lda	0.649022	0.634428	0.649146	0.643131	0.659287	0.647003
mini_bert	0.649326	0.633613	0.648955	0.642522	0.658777	0.646639
kmeans_bert	0.649341	0.632957	0.648852	0.642882	0.658735	0.646553
kmeans_mini_bert	0.649219	0.631721	0.649527	0.642484	0.659246	0.646439
Industry_Label_LDA	0.563016	0.589102	0.589102	0.589102	0.589102	0.583885
Combined	0.550118	0.592393	0.582919	0.583727	0.565279	0.574887
Vectorized_Industry_Label_comb	0.590911	0.572697	0.571172	0.570664	0.550256	0.571140
Vectorized_Industry_Label_lda	0.565422	0.551308	0.581718	0.589763	0.563058	0.570254
mini	0.552643	0.567416	0.568752	0.569809	0.553808	0.562486
LDA_NMF_Industry_Label	0.532018	0.562945	0.561087	0.571575	0.557748	0.557075
LDA_NMF	0.530905	0.563535	0.562295	0.571735	0.556625	0.557019
kmeans	0.553857	0.559899	0.553307	0.553225	0.548602	0.553778
LDA	0.583516	0.542863	0.536465	0.537990	0.534658	0.547098
LDA_TF_Industry_Labels	0.562338	0.543341	0.534493	0.535840	0.534537	0.542110
NMF	0.412419	0.590193	0.571241	0.578281	0.553858	0.541199
Industry_Labels_Only	0.425329	0.564149	0.565487	0.565487	0.565487	0.537188
NMF_TF_Industry_Labels	0.416674	0.566481	0.553737	0.570303	0.553661	0.532171
Vectorized	0.595868	0.461853	0.495777	0.490938	0.486384	0.506164
Vectorized_Industry_Label_nmf	0.569935	0.466448	0.498098	0.491066	0.499899	0.505089
Industry_Label_NMF	0.668946	0.462518	0.462518	0.462518	0.462518	0.503803

From the results above we found our best performing model was logistic regression which achieved an F1 score of .668 with the NMF industry labels feature set. But this feature set performed the worst on average across all the models. The next best performing model was catboost with an F1 score of .659 with BERT embeddings, LDA, and NMF industry labels combined. Looking at the average performance scores from all the models, it was obvious that the BERT feature sets performed the best, on average.

	Logistic	RandomForest	XGBoost	LightGBM	CatBoost	Set_Average
BERT_Industry_Label_nmf	0.649104	0.622542	0.667740	0.650240	0.663208	0.650567
BERT_Industry_Label	0.649374	0.622844	0.667170	0.650447	0.662725	0.650512
BERT_Industry_Label_Ida	0.649022	0.622538	0.667237	0.650319	0.662653	0.650354
bert	0.648980	0.623711	0.667012	0.649730	0.661946	0.650276
mini_bert	0.649326	0.621483	0.667057	0.650479	0.661937	0.650056
kmeans_bert	0.649341	0.619385	0.667606	0.650707	0.662290	0.649866
kmeans_mini_bert	0.649219	0.617054	0.667821	0.650503	0.662546	0.649428
Industry_Label_LDA	0.563016	0.639011	0.589102	0.589102	0.589102	0.593866
Vectorized_Industry_Label_Ida	0.565422	0.552484	0.580518	0.579737	0.580360	0.571704
Combined	0.550118	0.584012	0.561510	0.577360	0.566144	0.567829
Industry_Label_NMF	0.668946	0.432233	0.462518	0.462518	0.462518	0.497746

We saw minimal improvements from tuning hyperparameters across the board. Our average F1 increased by about .007 at a feature set average and by a maximum of .01 on a model basis. Our best model was still logistic regression with the Industry_Label_NMF feature set followed closely by XGBoost which had an F1 score of .6677 with BERT_Industry_Label_nmf along with all of the other feature sets that included BERT embeddings.

Conclusion and Further Study

In conclusion, we were able to find key insights into our data through our exploratory data analysis and NLP modeling we were able to achieve an F1 score of .668 through our best performing model, logistic regression. Random forest had the lowest performance of each of the models we evaluated and also had the longest training time. Going forward, we would stick to logistic regression, XGBoost, LightGBM, and Catboost.

Training time was a constraint for assessing our models on this dataset. If we had more time we would have liked to evaluate ensemble voting classifiers to see if we could get an improvement in our scores. We would also like to retry tuning our models with more trials.

Due to time constraints we had to use a lower number of trials, if training time were not a factor we could have seen better results from hyperparameter tuning.

References

About — Kickstarter. About - Kickstarter. (n.d.). <https://www.kickstarter.com/about>

Kaggle dataset: <https://www.kaggle.com/datasets/oscarvilla/kickstarter-nlp>